

Question number	Answer	Additional guidance	Mark
(a)(i)	An explanation that includes the following points: • {to amplify / to increase the sample / to increase the number of copies} (of DNA) (1) • so that there is {enough / more} to {analyse / use in profiling / use in gel electrophoresis} (1)	ACCEPT (DNA) samples are too small DO NOT ACCEPT to increase the length of DNA ACCEPT so will not be enough to analyse IGNORE easier / more accurate	(2)

Question number		Additional guidance	Mark
2(a)(ii)	A description that includes two of the following points: - to bind to the {DNA / primers} (1) - and line up the nucleotides along the DNA strands (1) - and form phosphodiester bonds between the nucleotides (on one strand) (1)	ACCEPT forms a (new) strand of DNA / adds nucleotides to the new strand / helps in formation of hydrogen bonds between base pairs / causes complementary base pairing	(2)

Question number	Answer	Mark
2(b)(i)	The correct answer is B (restriction enzyme) <i>A is incorrect because integrase joins viral DNA into host DNA</i> <i>C is incorrect because reverse transcriptase makes a DNA copy of RNA</i> <i>D is incorrect because RUBISCO is involved in the light-independent stages of photosynthesis</i>	(1)

Question number	Answer	Additional guidance	Mark
2(b)(ii)	An answer that includes three of the following points: - the (original) broccoli and the {genetically modified / clubroot resistant} broccoli OR the linseed and the {genetically modified / clubroot resistant} broccoli (1) - genetically modified plants to check that the resistance gene had been inserted (1) - the original plants to identify the broccoli bands (1) - the linseed to identify the resistance {genes / bands} (1)	NB Piece together ACCEPT bands with resistance gene ACCEPT idea that the linseed profile is compared to modified broccoli to find a {common / over-lapping} band to show successful modification = 3 marks	(3)